DESIGN REPORT 2

BAJA SAEINDIA

CHENNAI - 2026

TEAM NAME : DIRT DEMON RACING

TEAM ID :BSC26105

FACULTY ADVISOR : MR.S.NAVEEN KUMAR

TEAM CAPTAIN : SARAN .V

MODELON SIMULATION REPORT

1D SIMULATION AND PURPOSE

A 1D longitudinal vehicle dynamics simulation was developed to predict straight-line performance of the Baja SAE vehicle. The model represents the vehicle as a point mass and calculates acceleration based on the balance between tractive forces at the tires and resistive forces, including rolling resistance and aerodynamic drag.

The simulation is used to support data-driven drivetrain design decisions by evaluating the effects of gear ratios, CVT tuning, tire size, and vehicle mass on acceleration and tractive effort. This approach enables rapid comparison of design alternatives while maintaining sufficient accuracy for Baja SAE performance prediction.

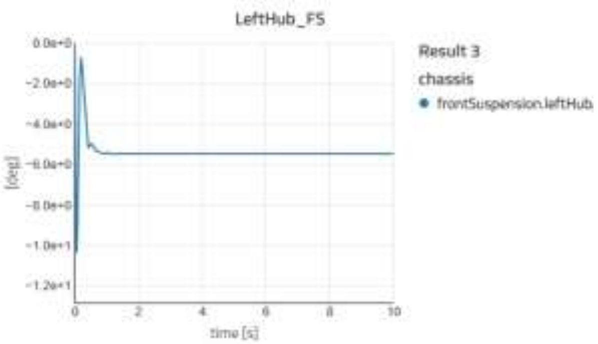
S.NO	TEST CASE
1	FLATPAD
2	FOURPOST
3	CORNEREXIT
4	ENGINETEST
5	MANUAL
6	ACCELERATION
7	BUMPTEST
8	HILLCLIMB
9	CLOSED LOOP

FLAT PAD

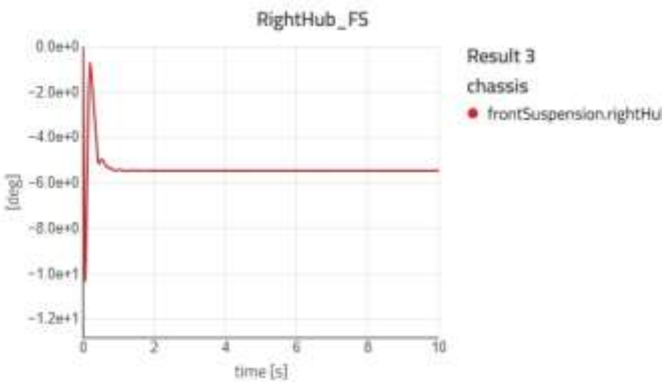
A Flat Pad Test is a vehicle evaluation method used to assess ride comfort and suspension performance. In this test, the vehicle is driven or excited on a flat surface that contains minor irregularities or disturbances. Even though the surface appears smooth, the small uneven inputs help engineers analyse how the vehicle responds dynamically.

TOE ANGLE

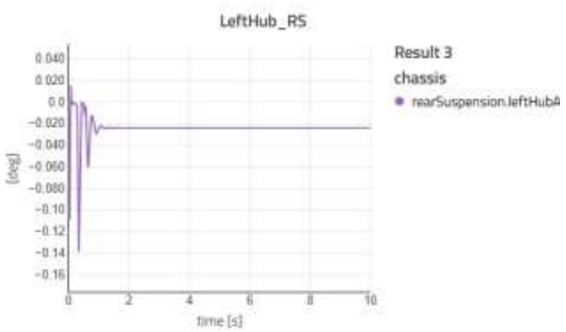
Front left



Front right



Rear left



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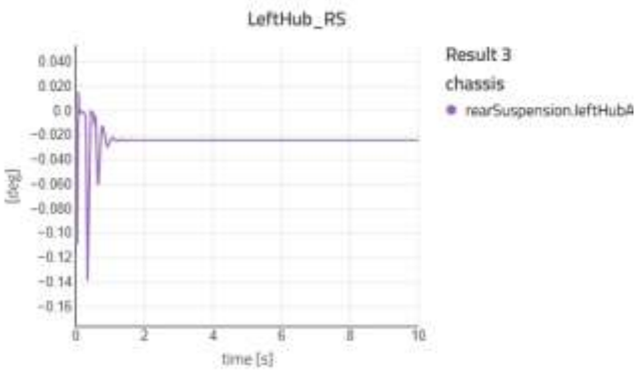
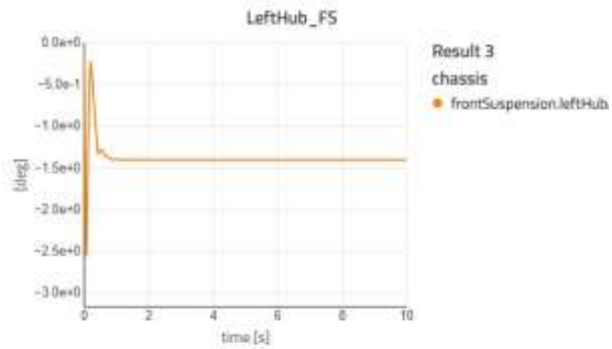


CHAMBER ANGLE

Frornt left



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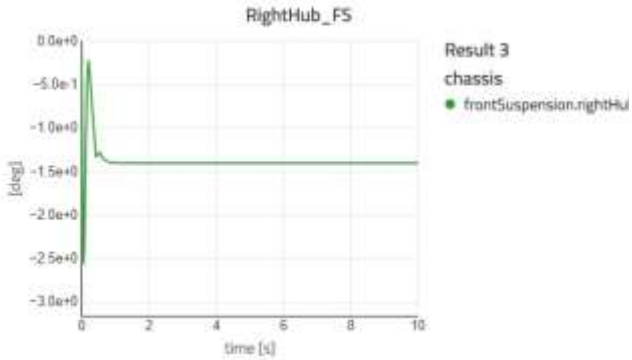

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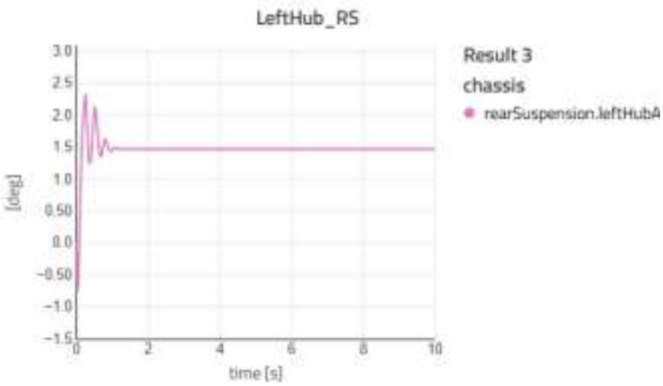
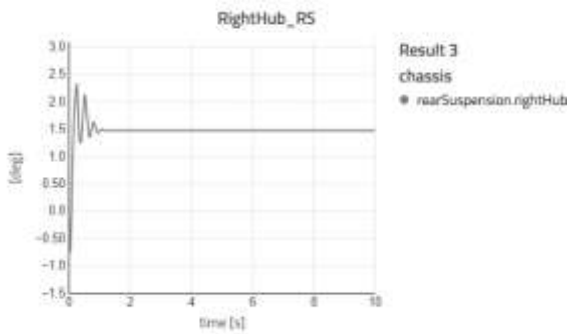


Rear right



Pass criteria

1. RMS vertical acceleration ≤ 0.3 g
2. peak vertical acceleration ≤ 0.6 g
3. No suspension bottoming out
4. No abnormal noise or vibration and Tyres remain in continuous contact with the pad

RESULT

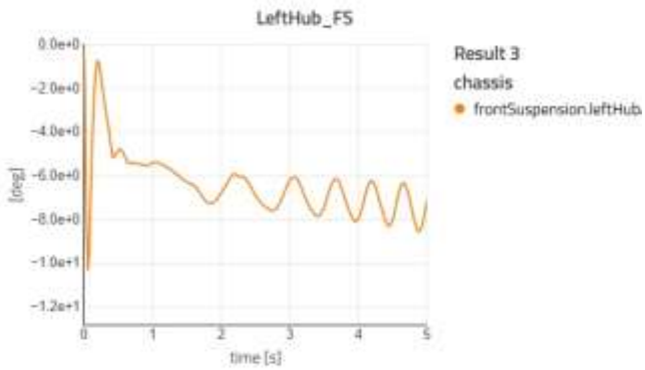
All test criteria is passed

FOUR POST TEST

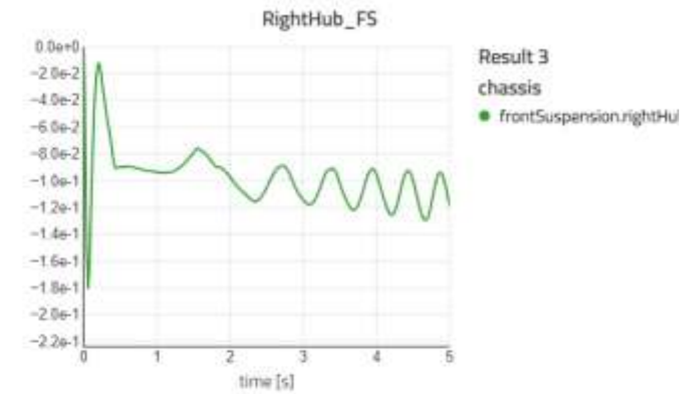
A Four Post Test is a vehicle durability and ride performance evaluation method carried out using a four-post-test rig. In this setup, each wheel of the vehicle is placed on an individual hydraulic actuator (post). These actuators simulate real road inputs by moving vertically according to predefined signals or recorded road data. This allows engineers to analyse vehicle behaviour under controlled laboratory conditions.

TOE ANGLE(FOUR POST)

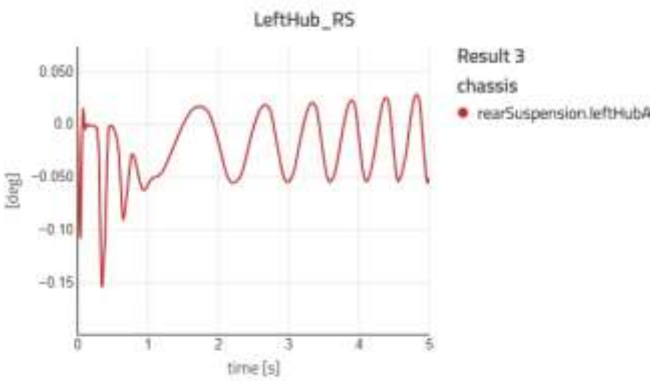
Front left



Front right



Rear left



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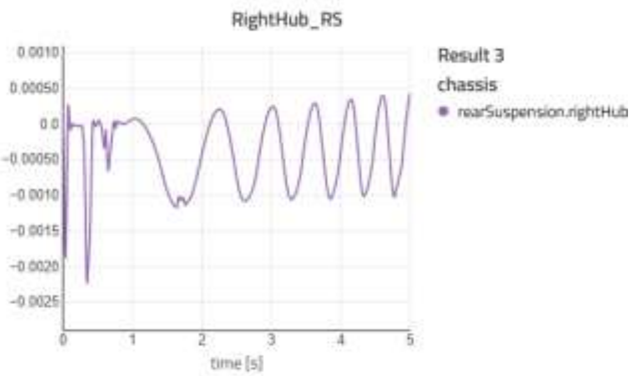
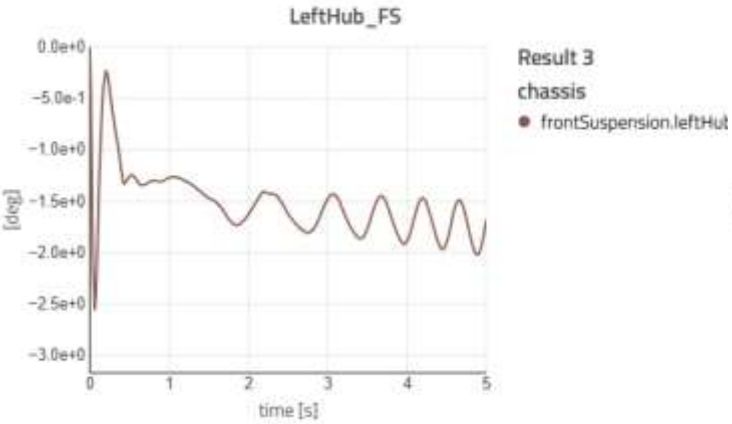


CHAMBER ANGLE(FOUR POST)

Frontleft



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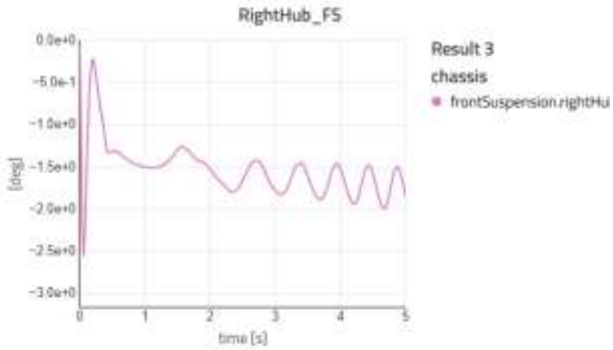
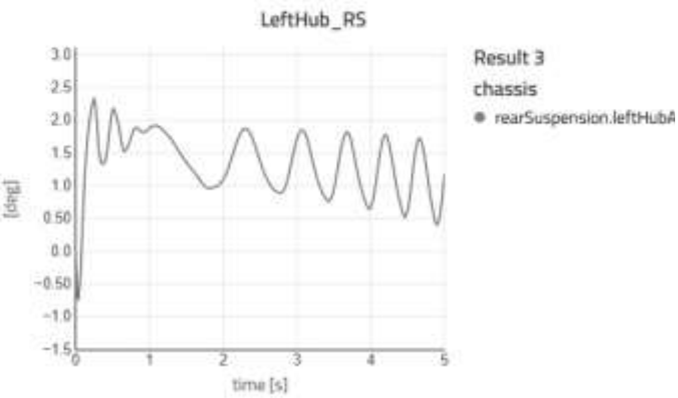


Rear left



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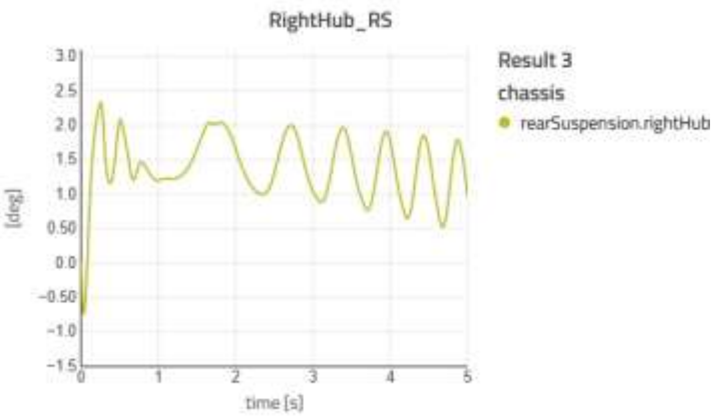
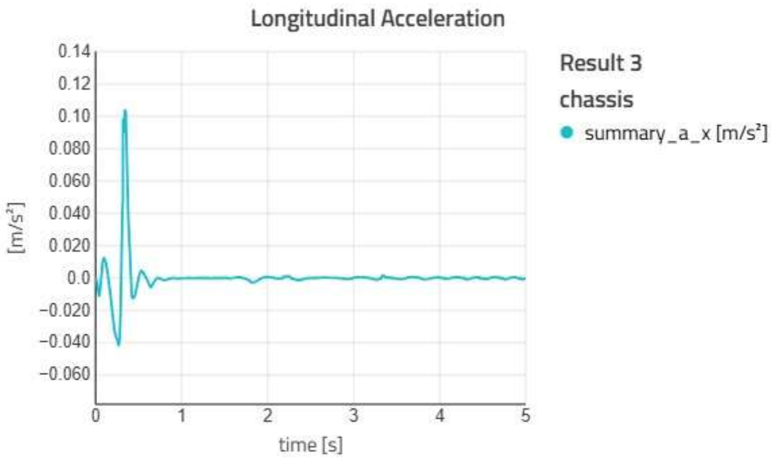
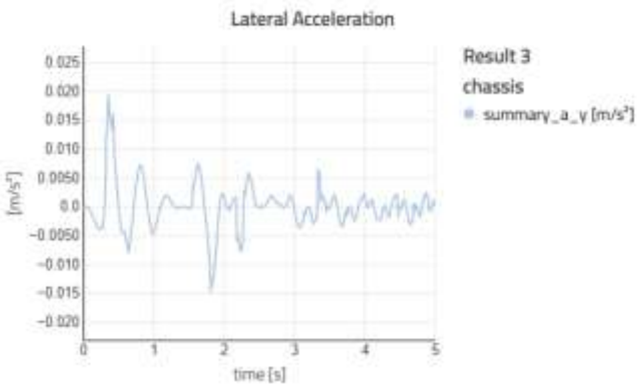


ACCELERATION

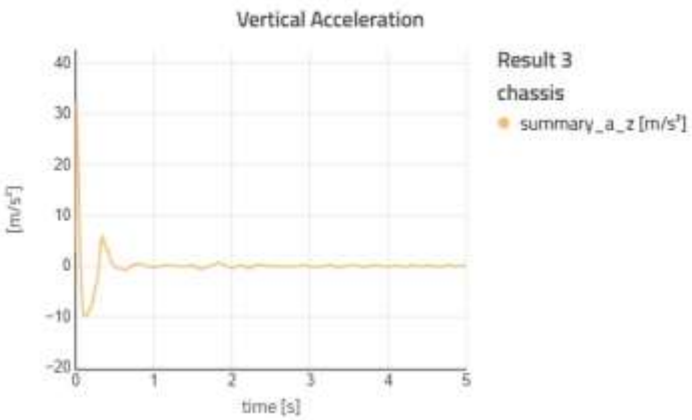
Longitudinal Acceleration



Lateral Acceleration

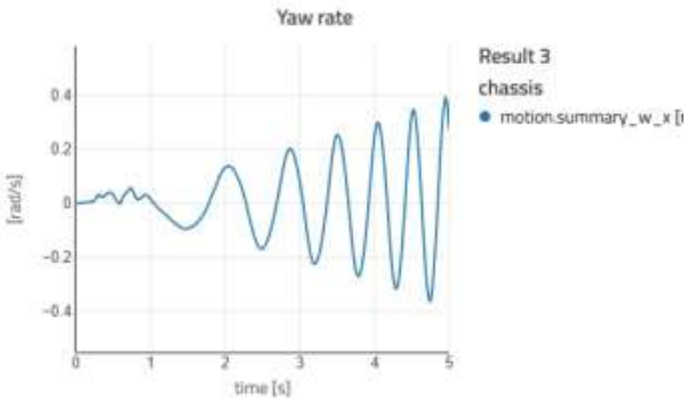


Vertical Acceleration

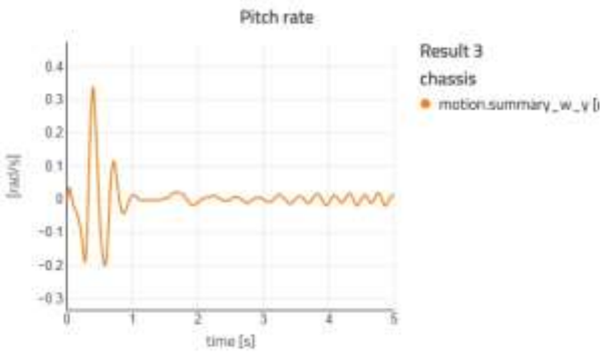


YAW,PITCH AND ROLE

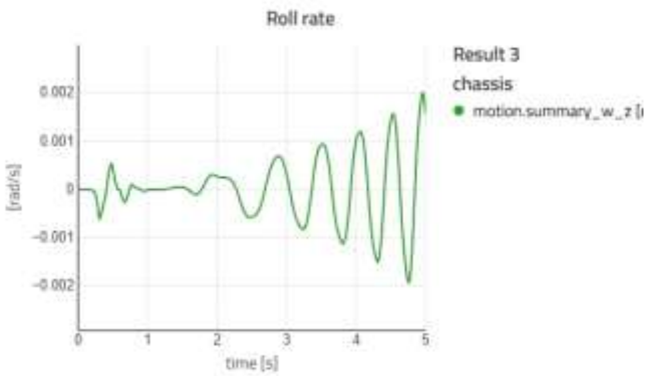
YAW RATE



PITCH RATE



ROLL RATE



Pass criteria

1. Body roll within ±3°
2. Pitch within ±2°
3. No structural stress failure
4. No excessive cabin vibration
5. Suspension travel remains within design limits

RESULT

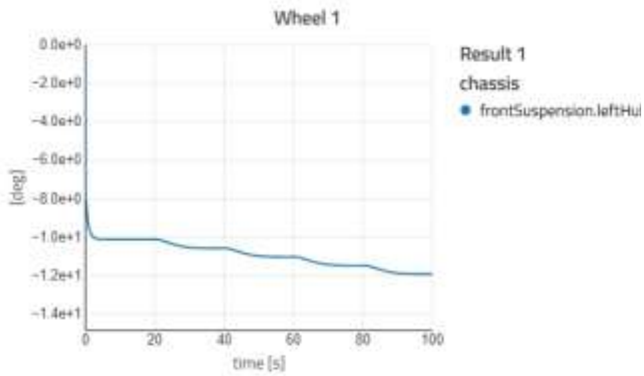
All test criteria is passed

CORNEREXIT

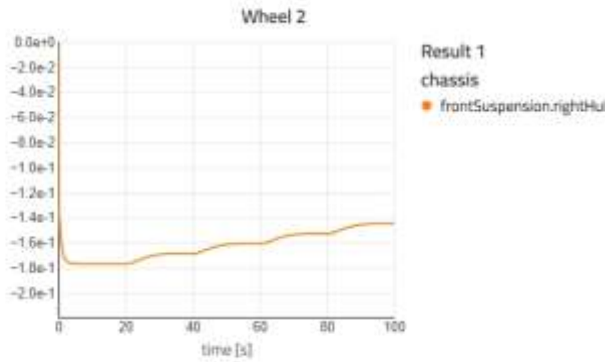
A Corner Exit Test is a vehicle dynamics evaluation performed to study the behaviour of a vehicle while accelerating out of a turn. This test focuses on traction, stability, steering response, and overall control when the driver applies throttle at the exit of a corner. It helps engineers assess how well the vehicle transfers power to the road during combined lateral and longitudinal loading.

TOE ANGLE

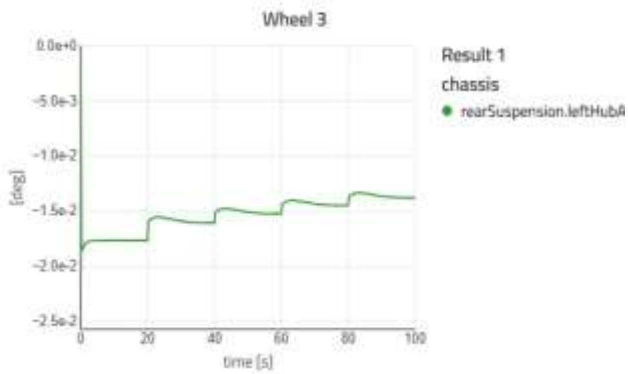
WHEEL 1



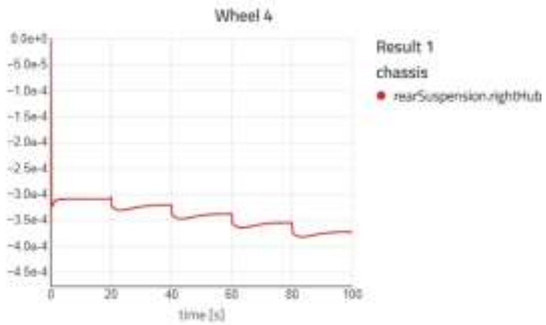
WHEEL 2



WHEEL 3

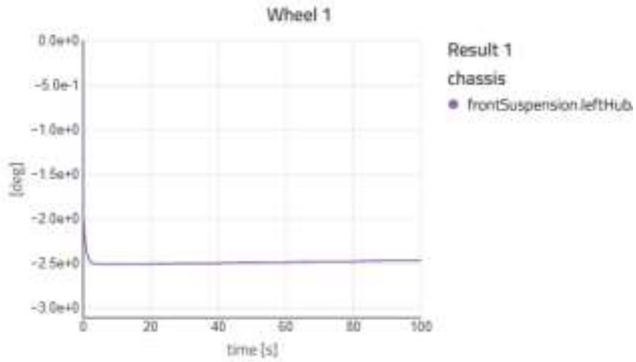


WHEEL 4

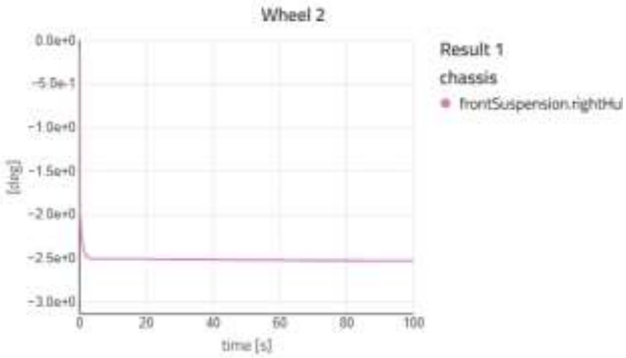


CHAMBER ANGLE

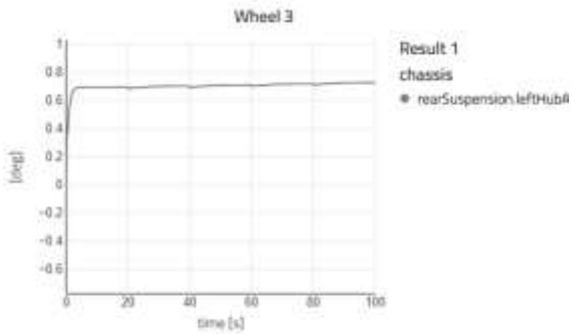
WHEEL 1



WHEEL 2



WHEEL 3



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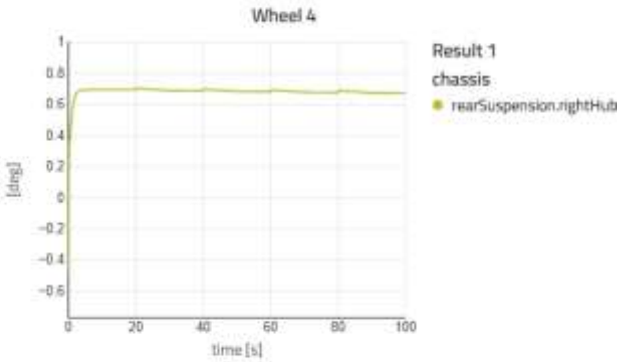
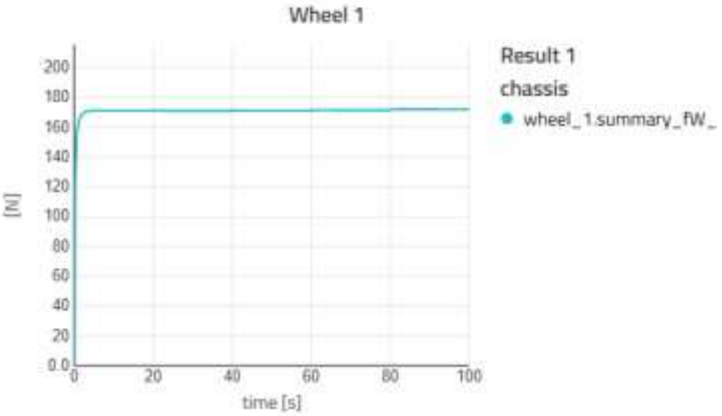
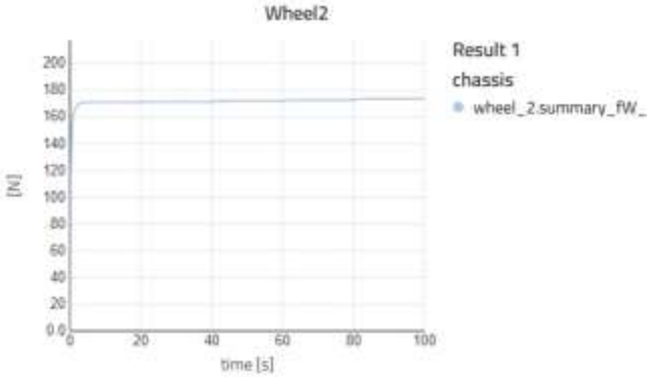


LONGITUDAL TIRE FORCE

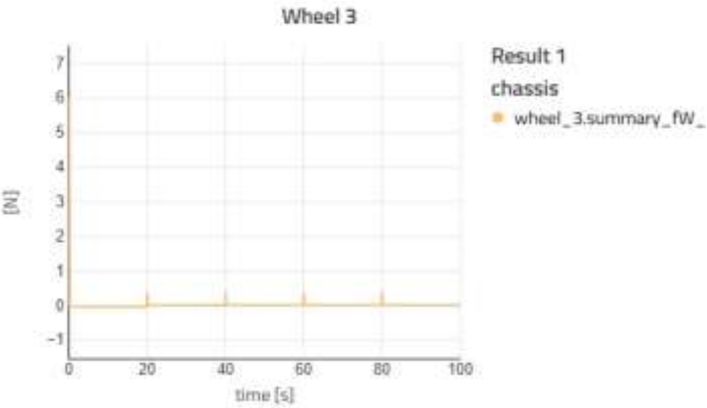
WHEEL 1



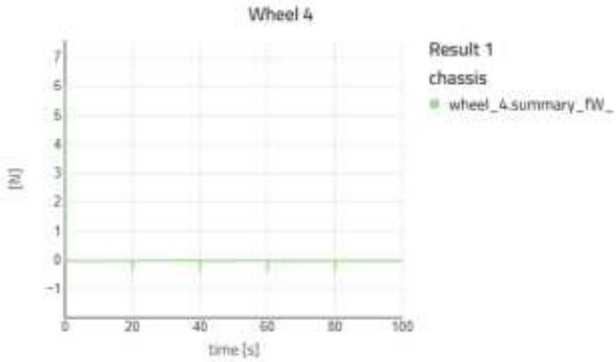
WHEEL 2



WHEEL 3



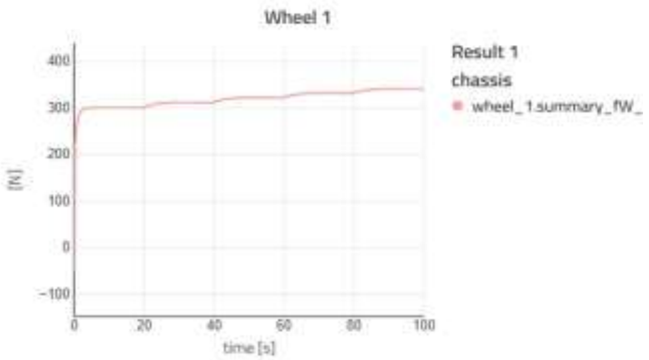
WHEEL 4

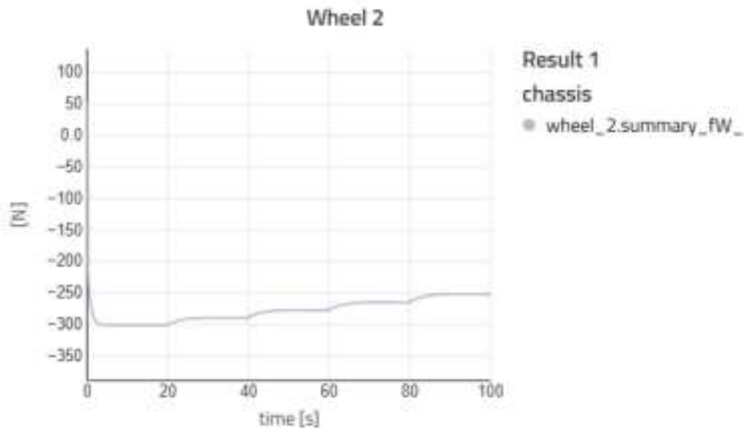


LATERAL TIRE FORCE

WHEEL 1

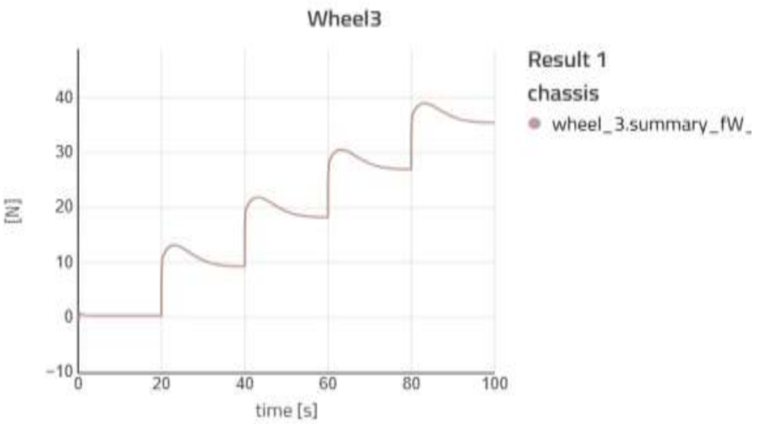
WHEEL 2

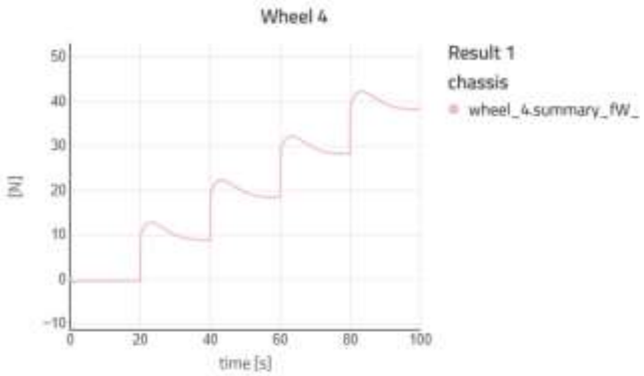




WHEEL 3

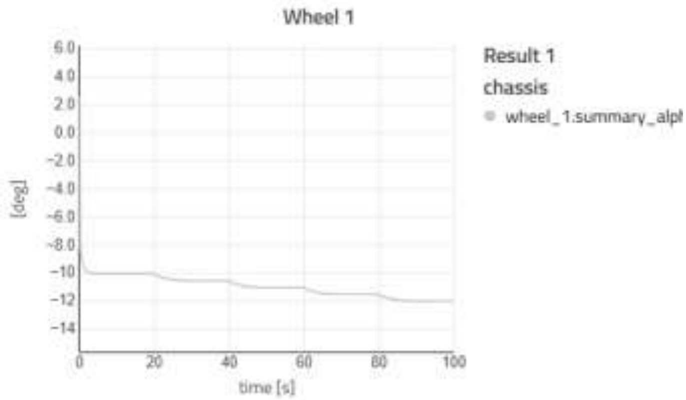
WHEEL 4



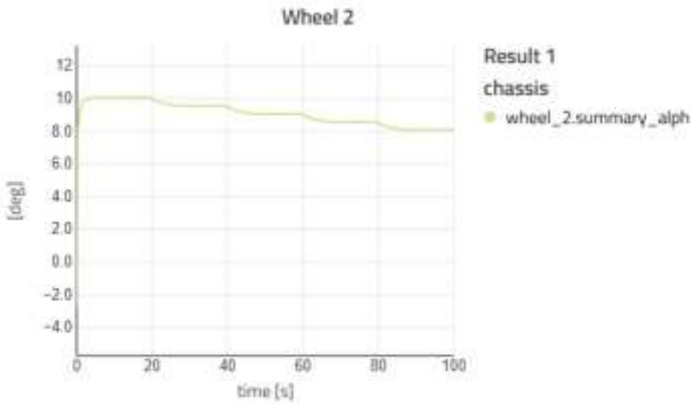


LATERAL SLIP ANGLE

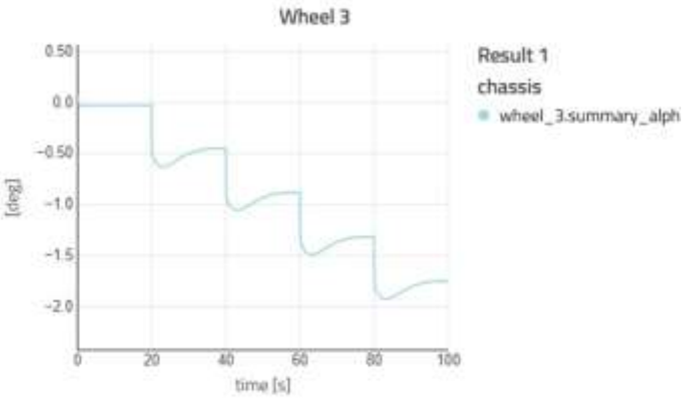
WHEEL 1



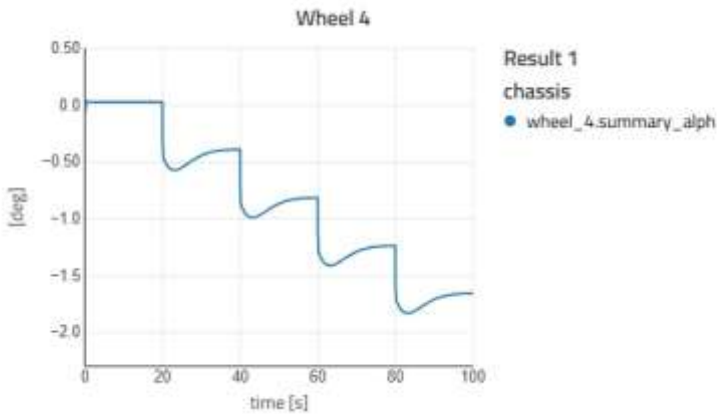
WHEEL 2



WHEEL 3

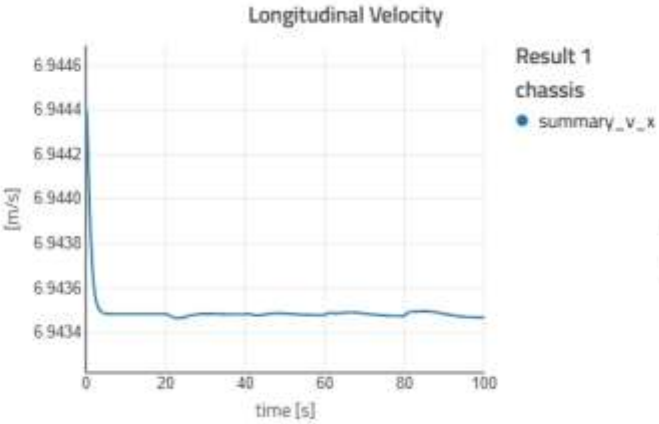


WHEEL 4

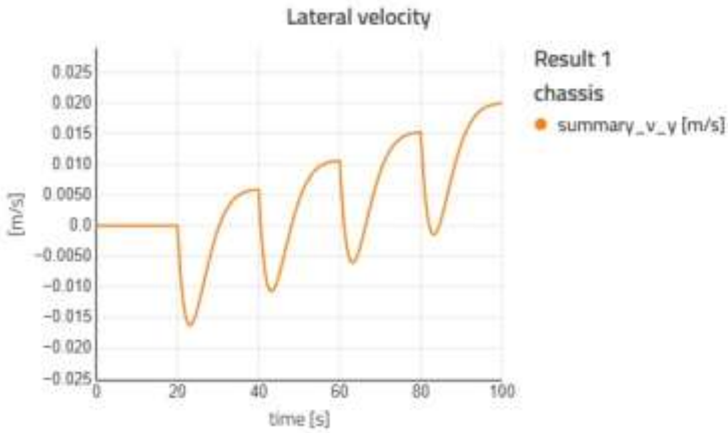


VELOCITY

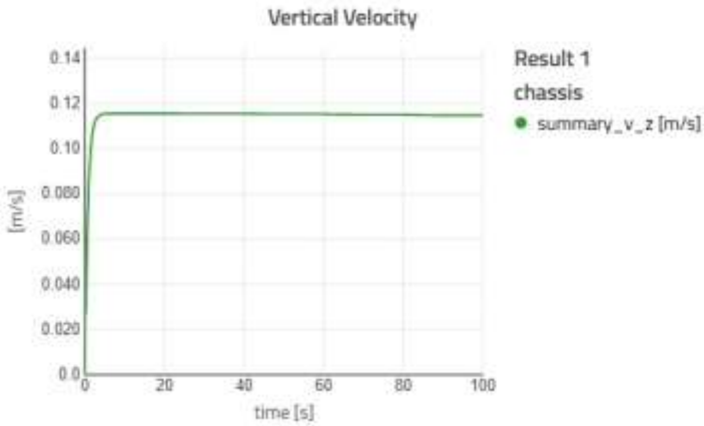
LONGITUDINAL VELOCITY



LATERAL VELOCITY

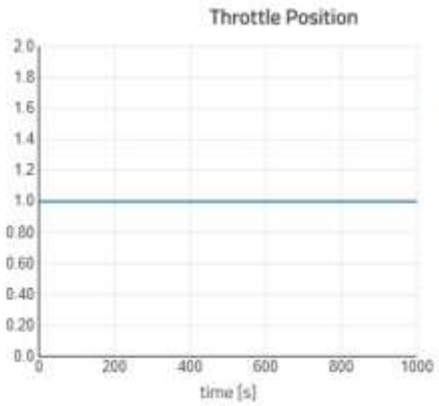


VERTICAL VELOCITY

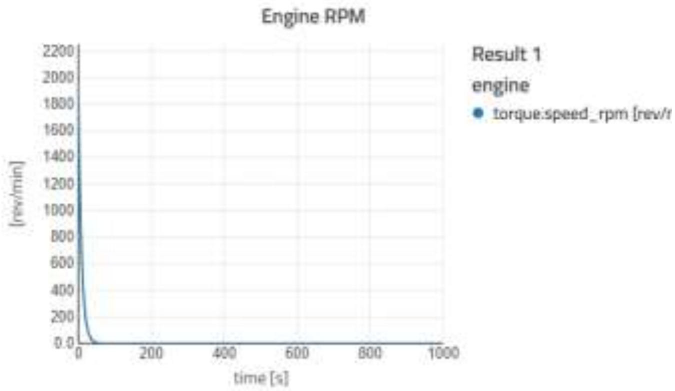


ENGINE TEST

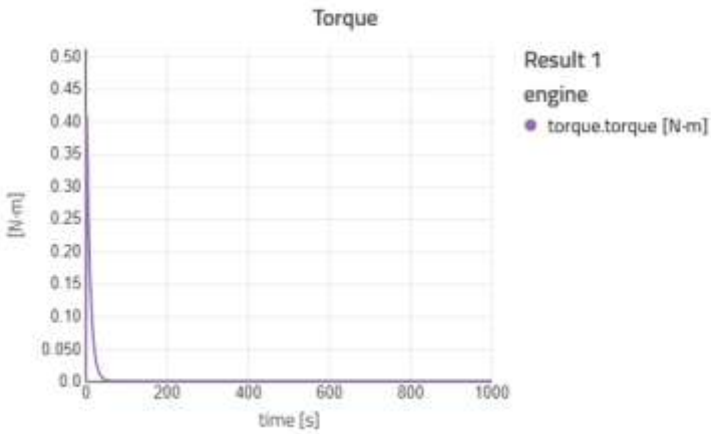
THROTTLE POSITION



ENGINE RPM



TORQUE



Pass criteria

1. Vehicle remains stable while exiting the corner
2. No sudden loss of control or skidding
3. Steering response is smooth and predictable
4. Tyre slip is within acceptable limit (not excessive)

Driver can maintain the lane easily

RESULT

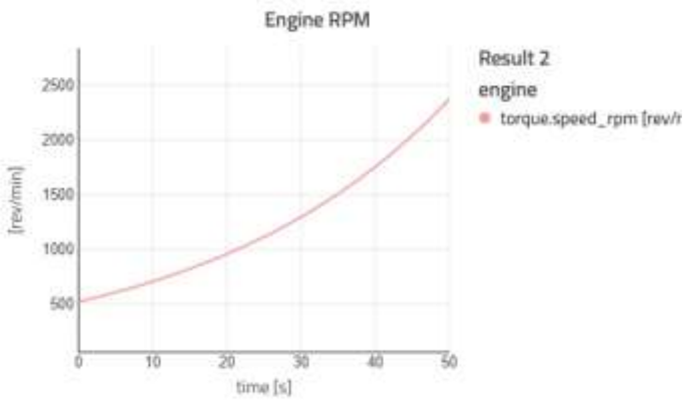
All test criteria is passed

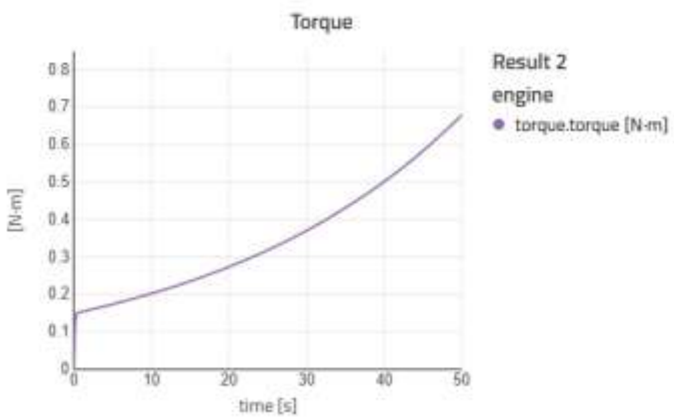
MANUAL

A Manual Transmission Test is conducted to evaluate the performance, durability, and shift quality of a vehicle equipped with a manual gearbox. This test focuses on clutch operation, gear engagement, shift smoothness, and overall drivability under different driving conditions.

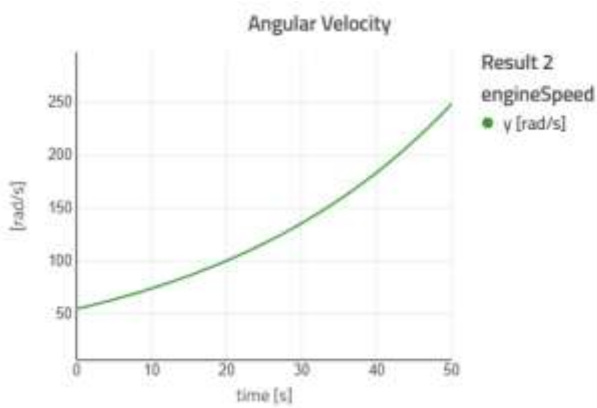
MANUAL ENGINE

ENGINE RPM **TORQUE**



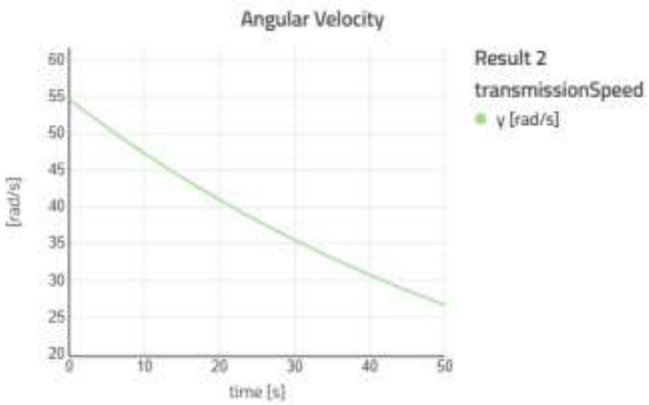


ANGULAR VELOCITY

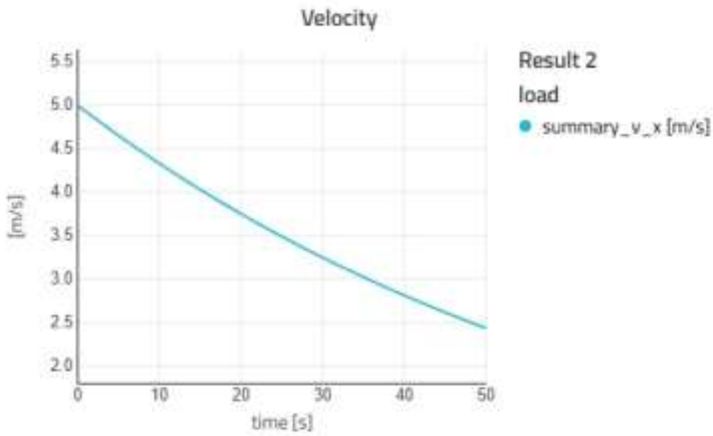


MANUAL (PERFORMANCE)

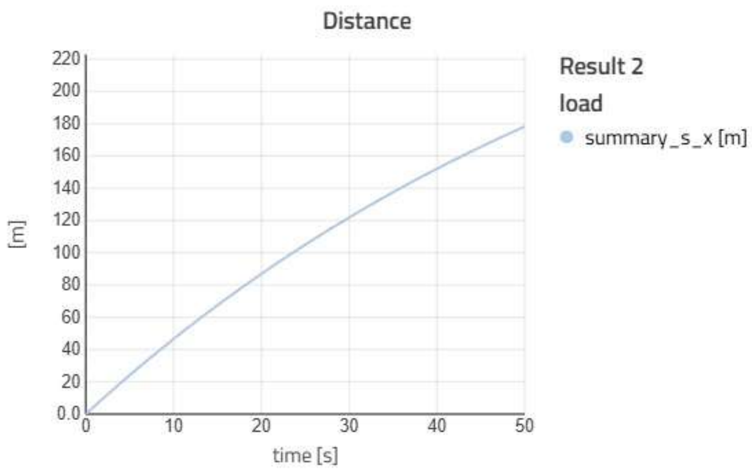
ANGULAR VELOCITY



VELOCITY



DISTANCE



Pass criteria

1. Engine does not stall
2. Steering remains stable and predictable
3. Braking distance is within acceptable limit
4. Vehicle feels easy to control for the driver

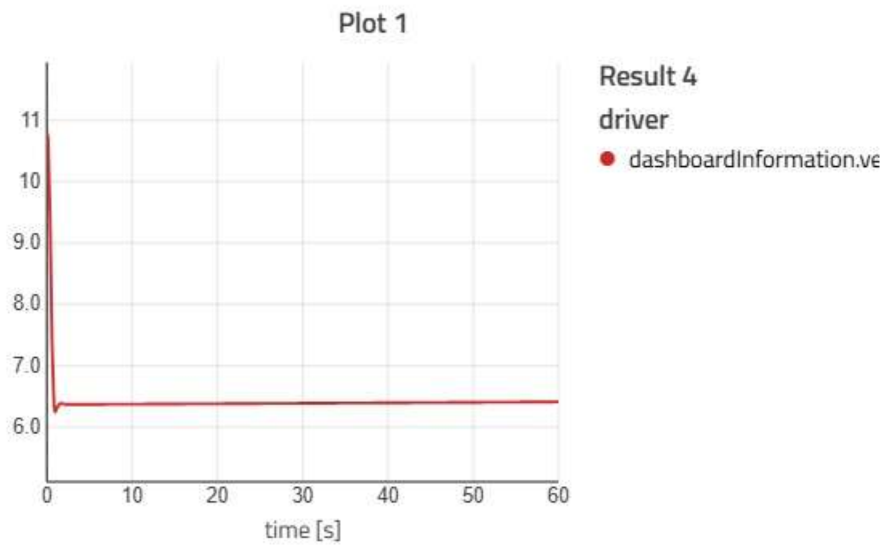
RESULT

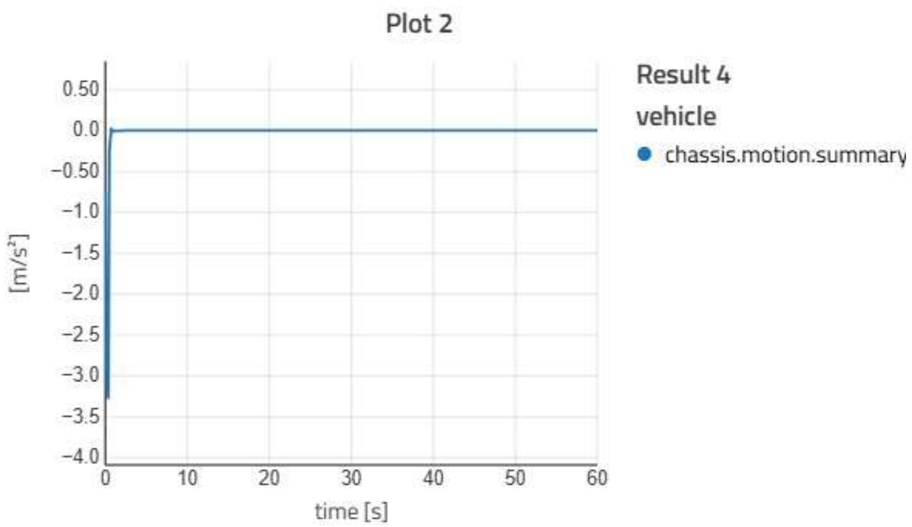
All test criteria is passed

ACCELERATION

An Acceleration Test is performed to measure how quickly a vehicle increases its speed from a standstill or while in motion. This test evaluates engine performance, drivetrain efficiency, throttle response, and overall vehicle power delivery.

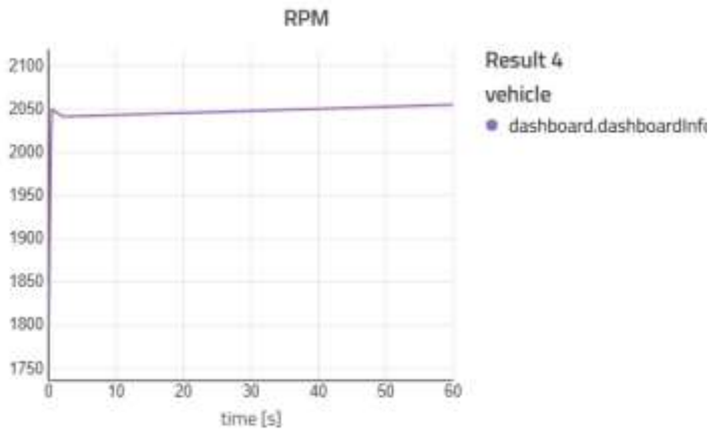
ACCELERATION (SPEED)



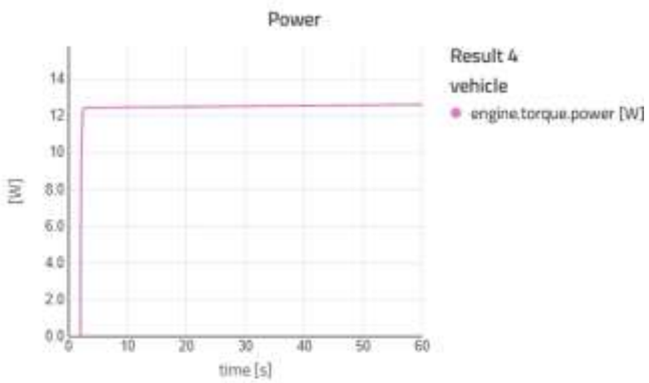


ACCELERATION (ENGINE)

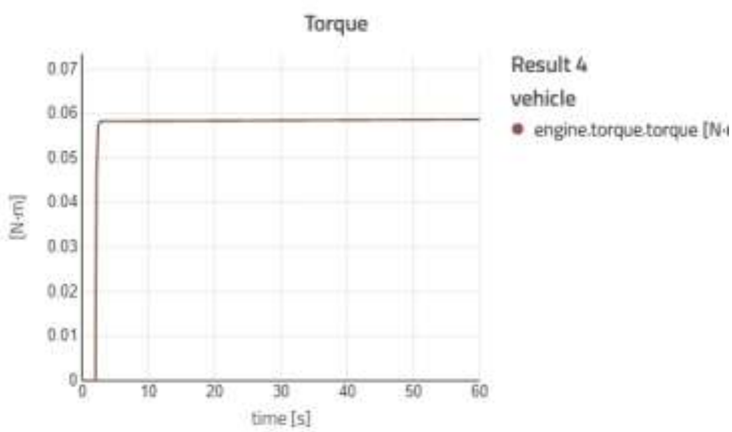
RPM



POWER



TORQUE



Pass criteria

1. 0–60 km/h is achieved within the target time (e.g., ≤ 8 seconds, based on your specification)
2. No excessive wheel spin (slip within acceptable limit)
3. No strong vibration in engine or drivetrain
4. Engine temperature remains stable during acceleration
5. Vehicle accelerates smoothly without jerks

RESULT

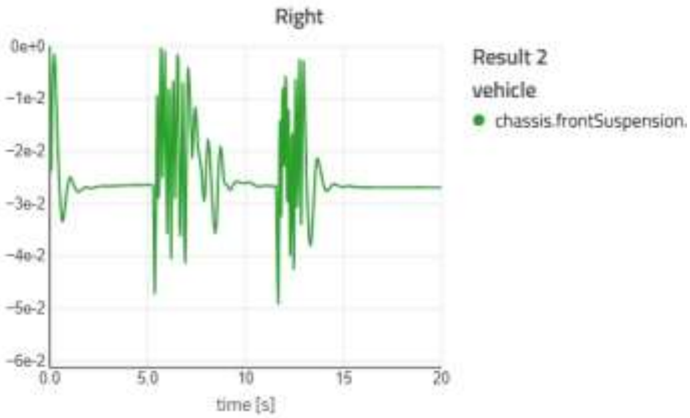
All test criteria is passed

BUMP TEST

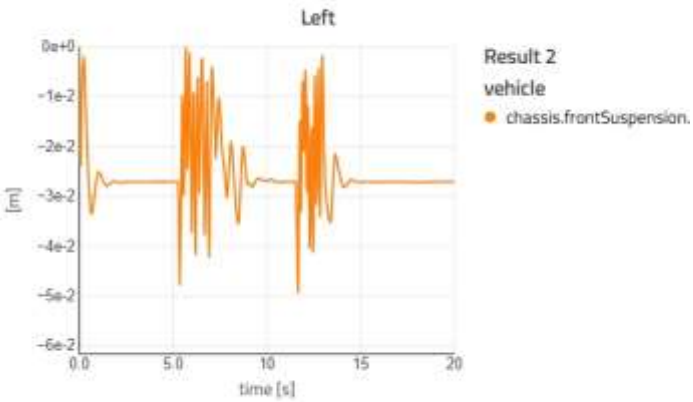
A Bump Test is a vehicle simulation or experimental test used to study how a vehicle responds when it passes over a sudden road irregularity such as a speed breaker, pothole, or raised surface. In this test, the vehicle is driven over a bump at a controlled speed to analyse its dynamic behaviour and suspension performance.

BUMP TEST (FRONT SUSPENSION)

RIGHT

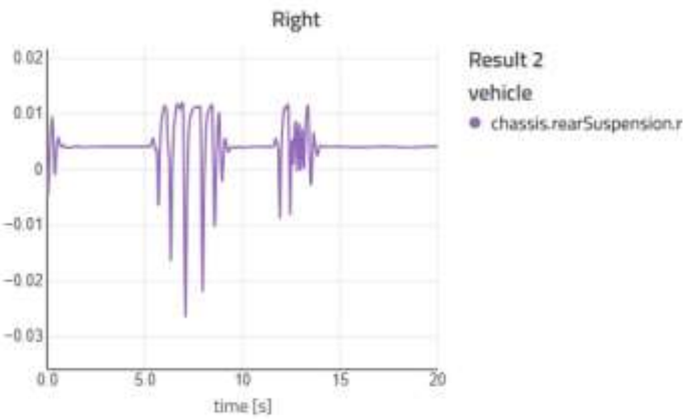


LEFT

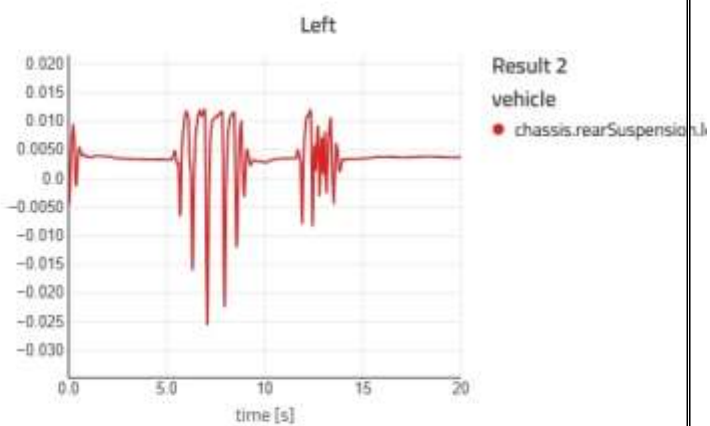


BUMP TEST (REAR SUSPENSION)

RIGHT



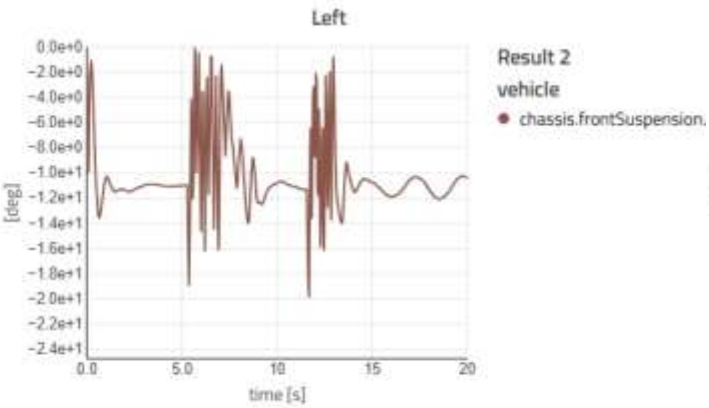
LEFT

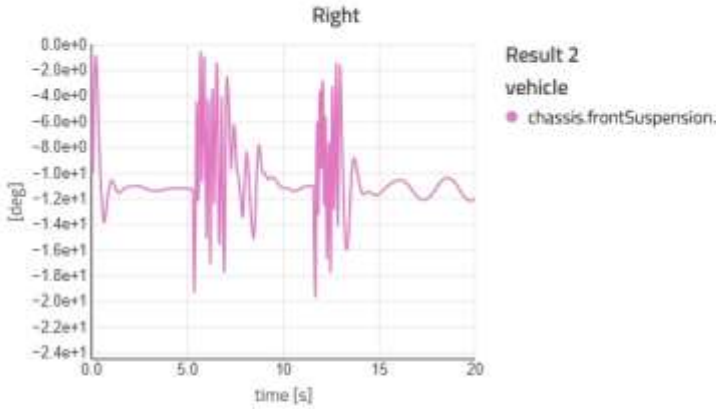


BUMP TEST (TOE)

FRONT

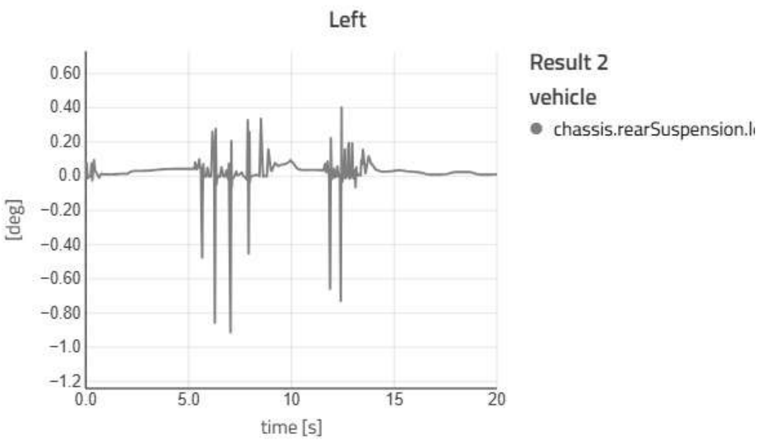
LEFT **RIGHT**

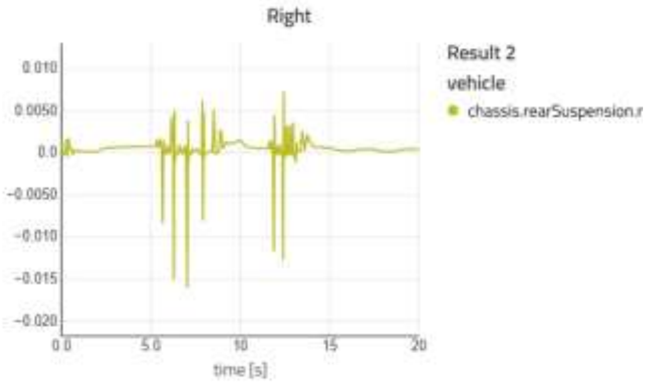




REAR

LEFT **RIGHT**

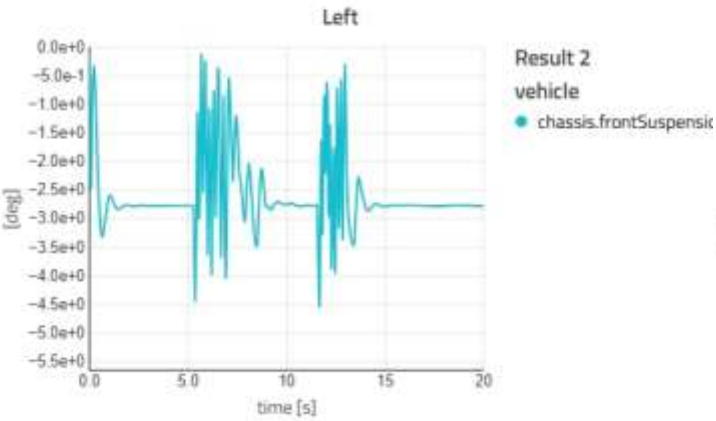


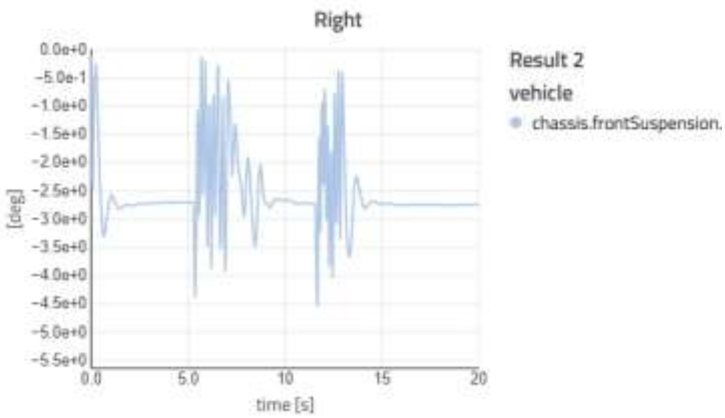


BUMP TEST (CHAMBER)

FRONT

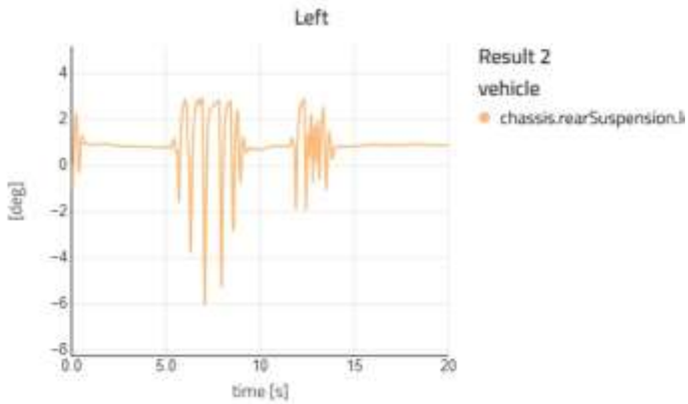
LEFT **RIGHT**

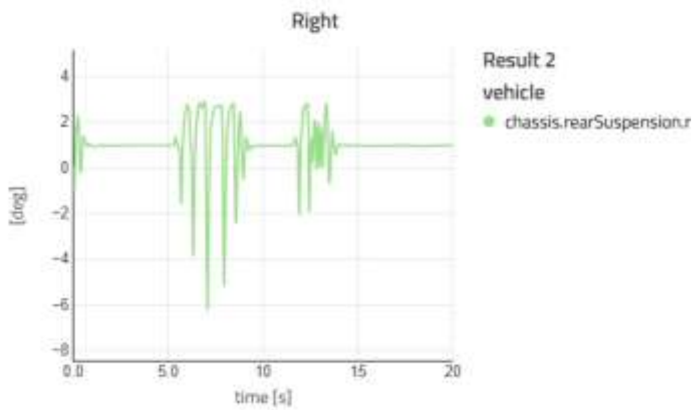




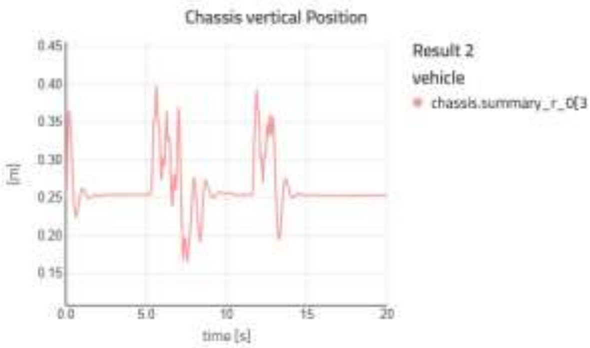
REAR

LEFT **RIGHT**

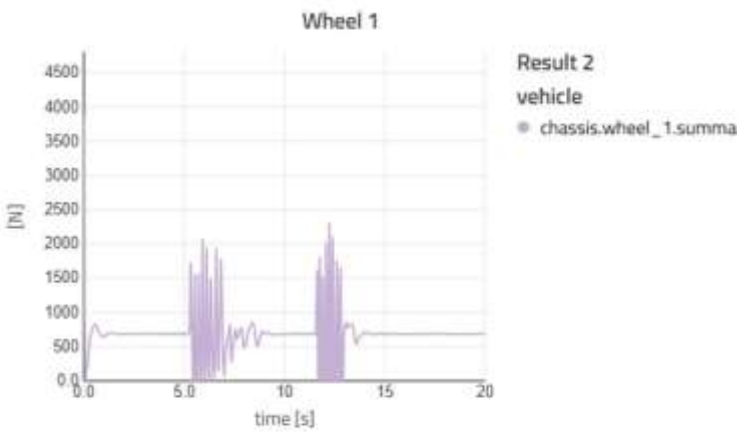


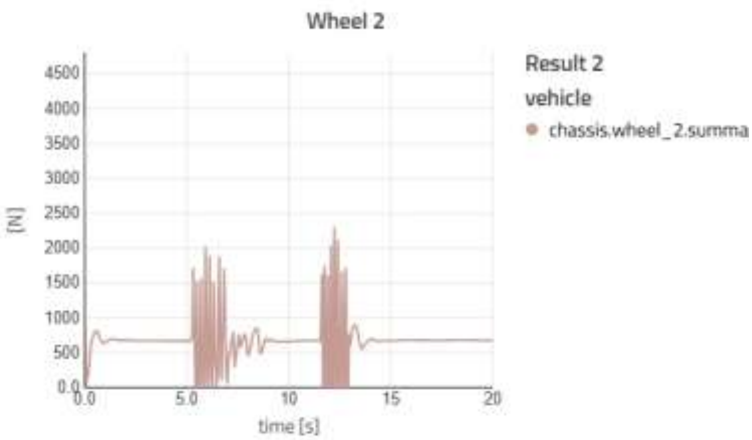


BUMP TEST (CHASSIS POSITIONS)

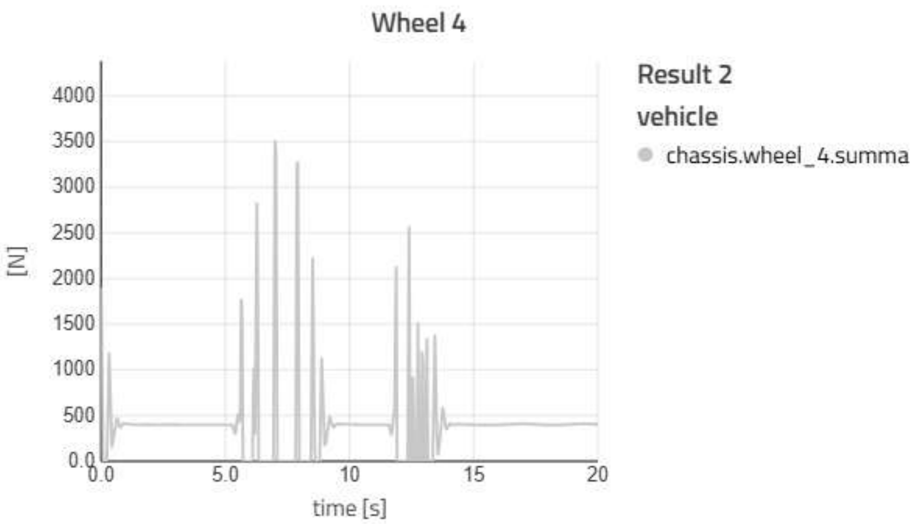


BUMP TEST (VERTICAL FORCES)






Wheel 3
Result 2
vehicle
chassis.wheel_3.summa
[N]
time [s]


Wheel 4
Result 2
vehicle
chassis.wheel_4.summa
[N]
time [s]

Pass criteria

1. No suspension bottoming out
2. Vehicle does not bounce excessively
3. Tyres maintain contact with the ground
4. Vehicle body settles within 2 seconds after the bump
5. No rattling noise or component damage

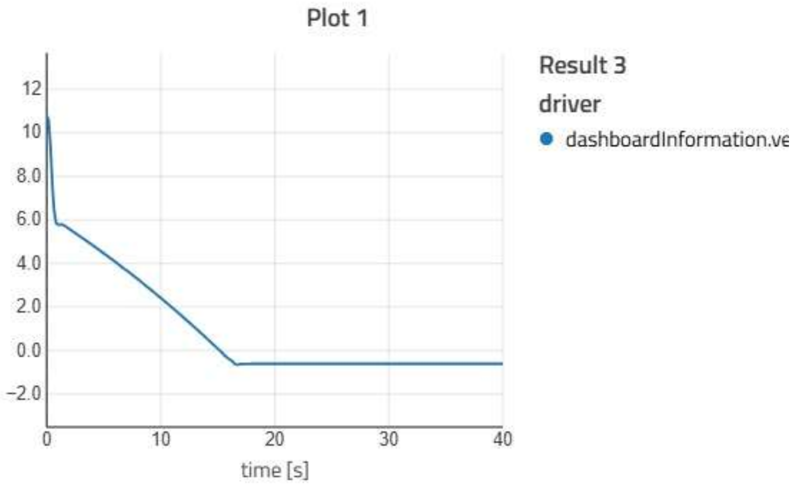
RESULT

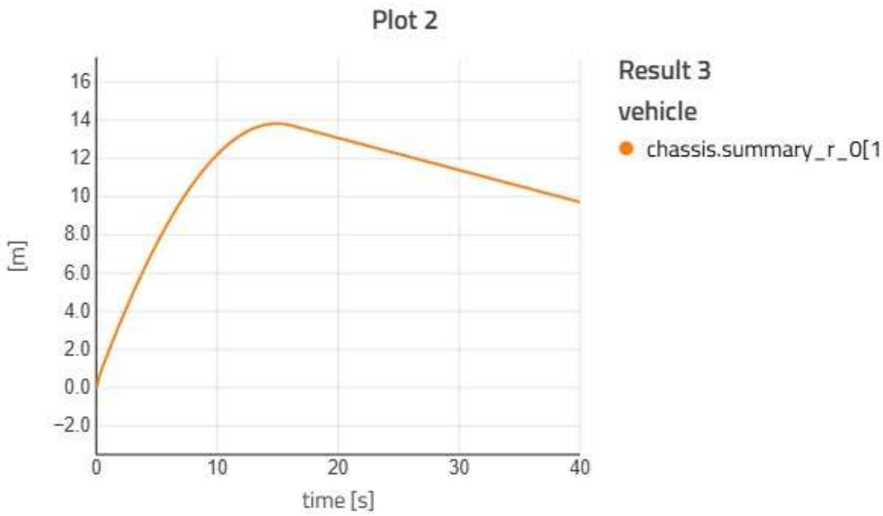
All test criteria is passed

HILLCLIMB

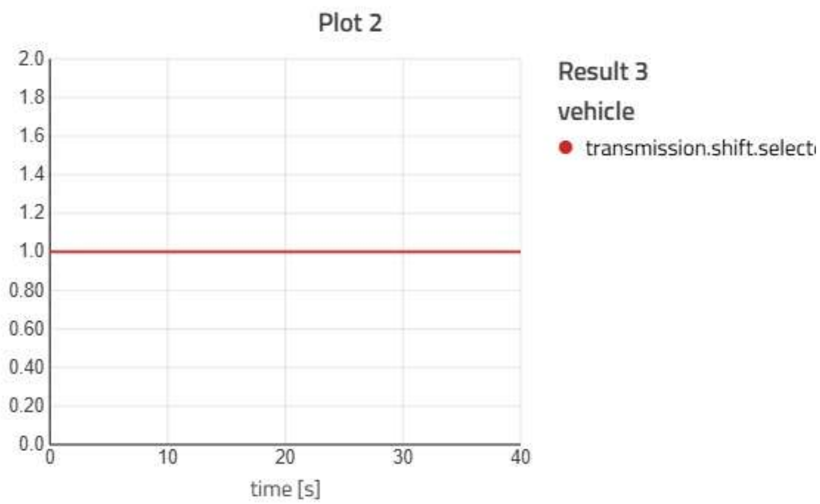
A Hill Climb Test is a vehicle simulation or experimental test used to evaluate how well a vehicle can travel uphill on an inclined surface. In this test, the vehicle is driven on a slope with a specified gradient to assess its climbing ability, traction, and stability.

HILLCLIMB (SPEED AND DISTANCE)



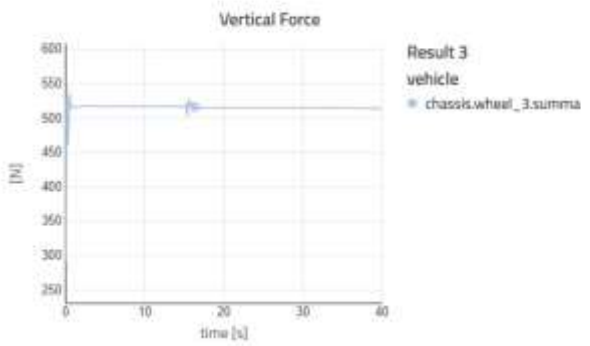


HILLCLIMB (POWER TRAIN)

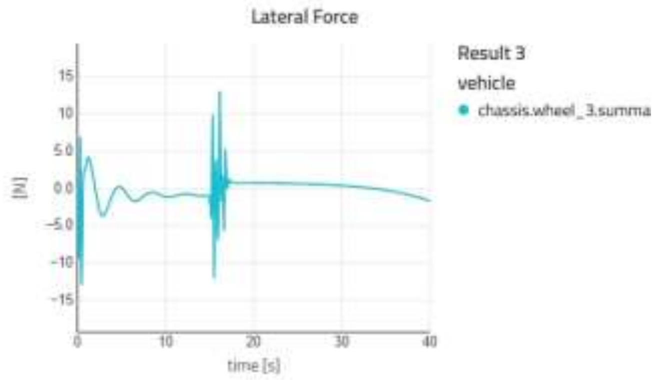


HILLCLIMB (TIRE FORE)

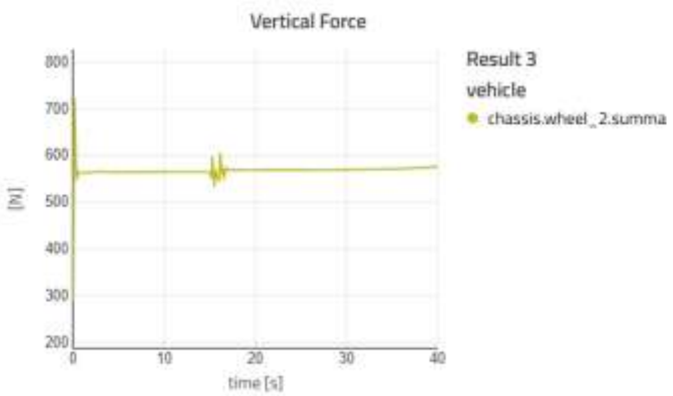
VERTICAL FORCE



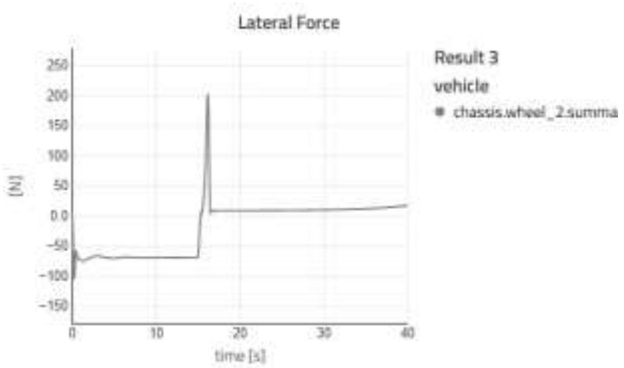
LATERAL FORCE



VERTICAL FORCE

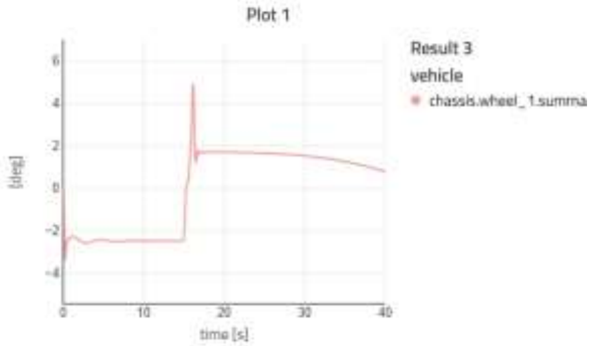


LATERAL FORCE

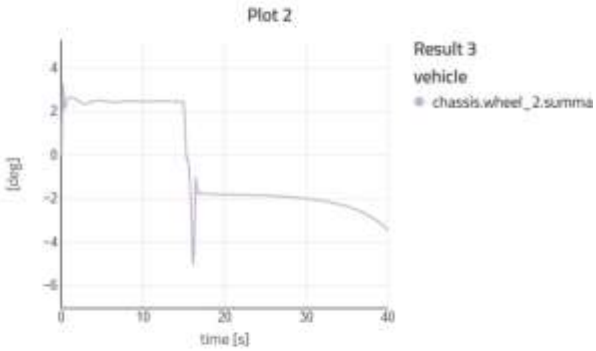


HILLCLIMB (LATERAL SLIP)

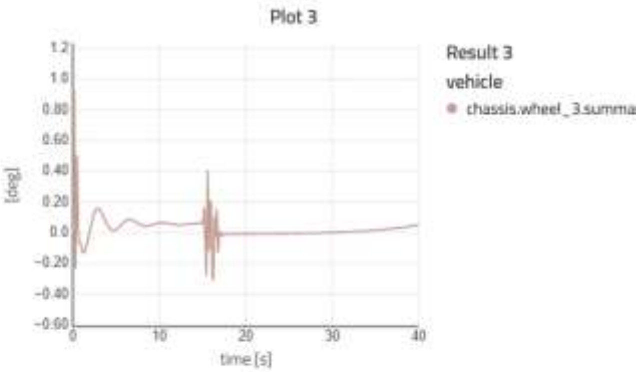
PLOT 1



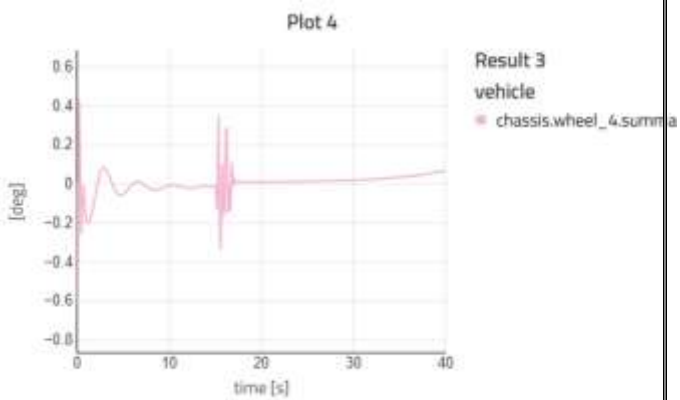
PLOT2



PLOT 3



PLOT4



Passing Criteria

1. Vehicle climbs a 15–20% gradient without rolling back
2. Engine does not overheat during the climb
3. Brakes can hold the vehicle on the slope
4. No excessive wheel slip occurs
5. Vehicle reaches the top smoothly without stalling

RESULT

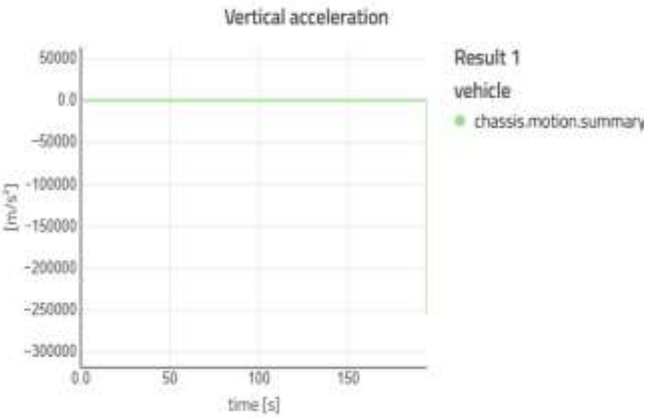
All test criteria is passed

CLOSEDLOOPDRIVER

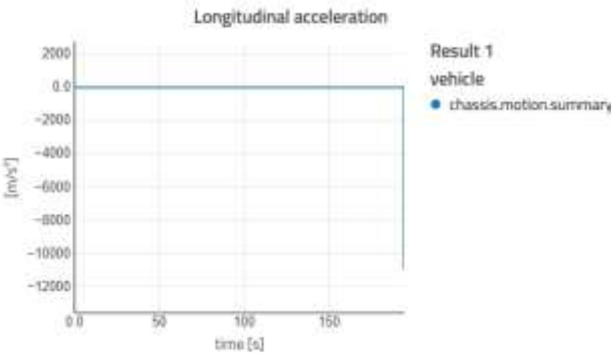
A Closed-Loop Driver Test is a vehicle simulation or control-based test in which a virtual or automated driver continuously adjusts steering, acceleration, and braking based on real-time feedback from the vehicle and the road. In this test, the driver system reacts automatically to keep the vehicle on a desired path or lane.

CLOSEDLOOP(ACCELERATION)

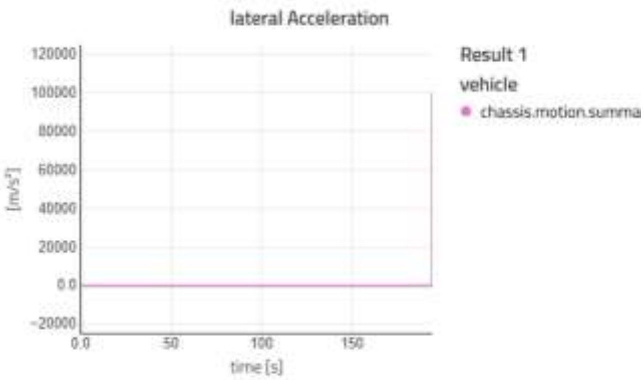
VERTICAL ACCELERATION



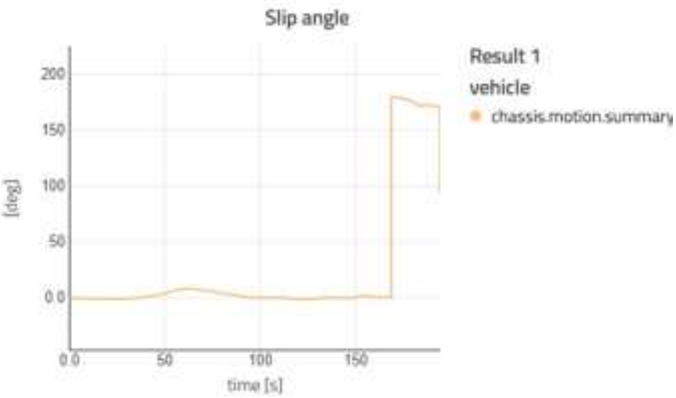
LONGITUDINAL ACCELERATION



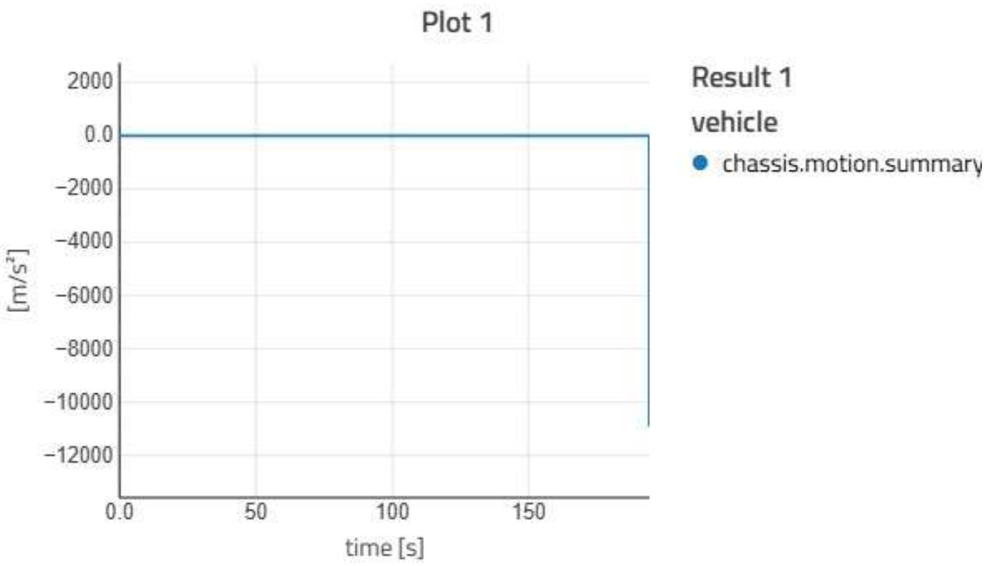
LATERAL ACCELERATION



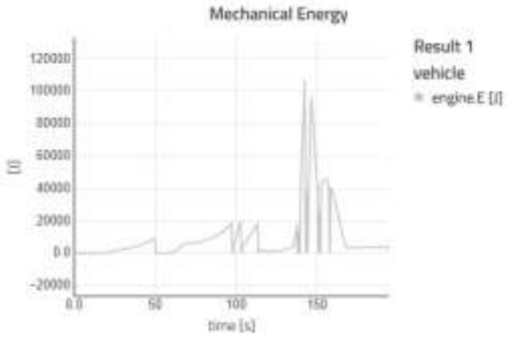
SLIP ANGLE

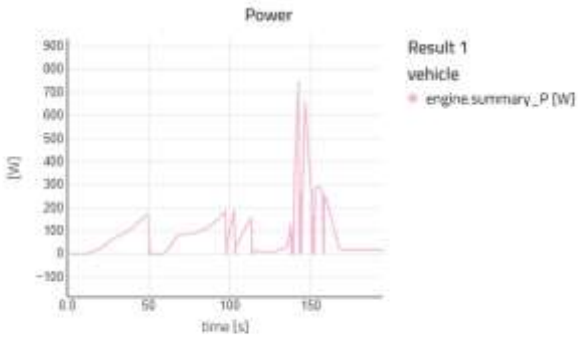


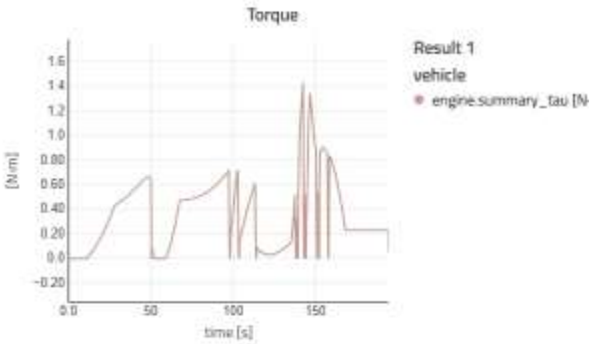
CLOSED LOOP (SPEED)

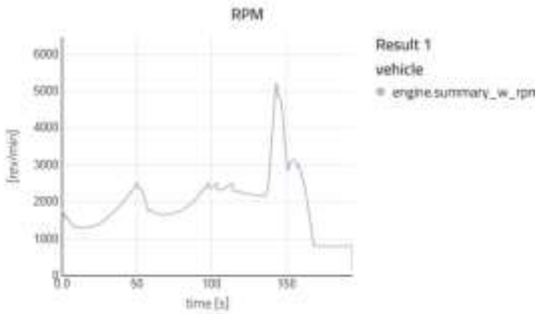


CLOSED LOOP (ENGINE DRIVE)

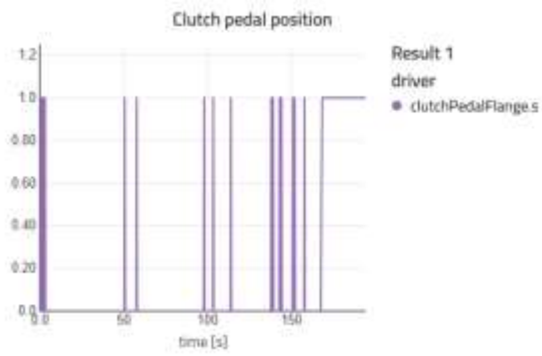


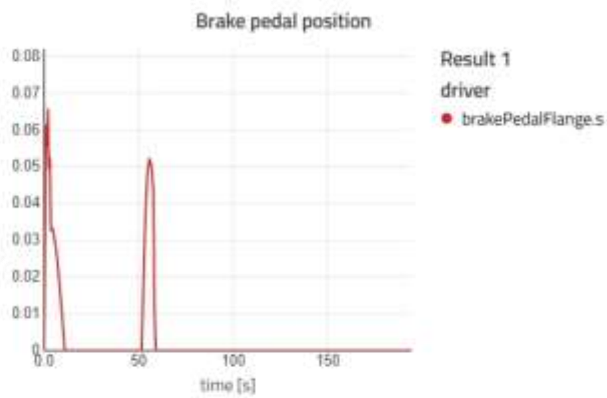


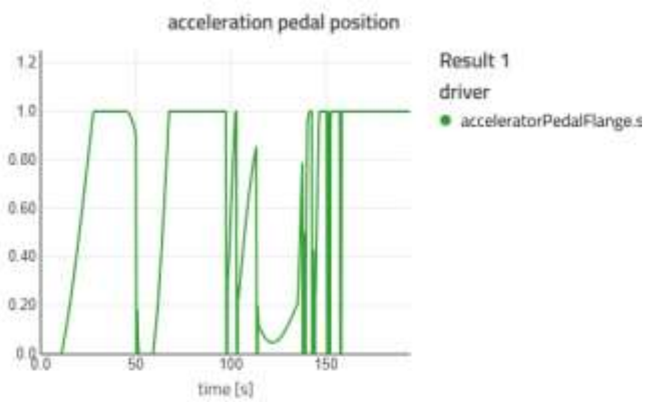




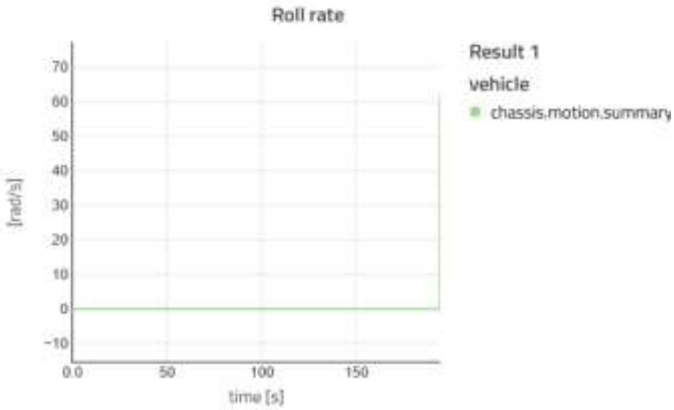
CLOSED LOOP (PEDAL POSITION)

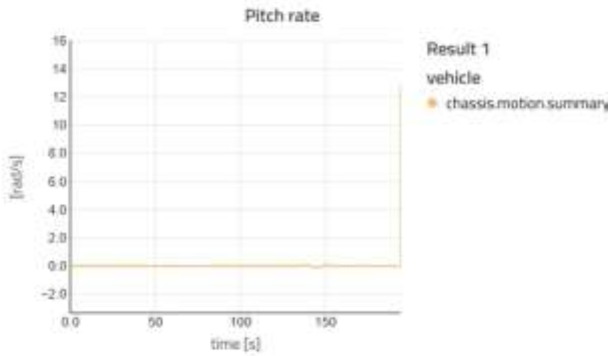


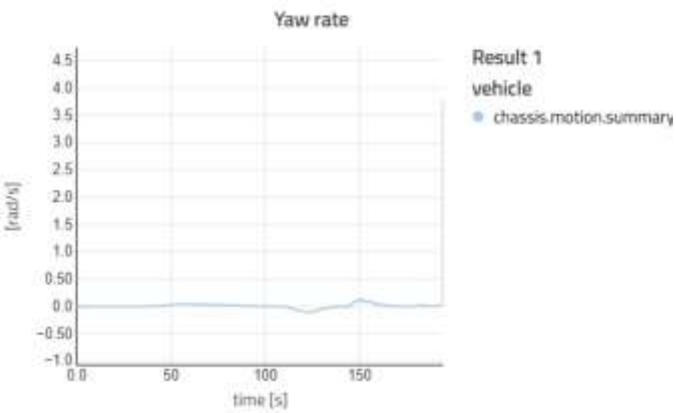




CLOSED LOOP(YAW,PITCH,ROLL)







Pass criteria

1. Path tracking error is within acceptable limit (≈ ≤ 0.2 m)
2. Steering response is smooth without sudden jerks
3. No oscillation or hunting behaviour in steering
4. Vehicle speed remains stable
5. Lane keeping is maintained throughout the test

RESULT

All test criteria is passed

ENGINE TORQUE TABLE

						Throttle position						
	0	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
	800	0	0.023	0.046	0.069	0.091	0.114	0.137	0.16	0.183	0.206	0.229
	1000	0	0.029	0.057	0.086	0.114	0.143	0.171	0.2	0.229	0.257	0.286
RPM	1200	0	0.034	0.069	0.103	0.137	0.171	0.206	0.24	0.274	0.309	0.343
	1500	0	0.043	0.086	0.129	0.171	0.214	0.257	0.3	0.343	0.386	0.429
	1800	0	0.051	0.103	0.154	0.206	0.257	0.309	0.36	0.411	0.463	0.514
	2000	0	0.057	0.114	0.171	0.229	0.286	0.343	0.4	0.457	0.514	0.571
	2200	0	0.063	0.126	0.189	0.251	0.314	0.377	0.44	0.503	0.566	0.629
	2500	0	0.071	0.143	0.214	0.286	0.357	0.429	0.5	0.571	0.643	0.714
	2800	0	0.08	0.16	0.24	0.32	0.4	0.48	0.56	0.64	0.72	0.8
	3000	0	0.086	0.171	0.257	0.343	0.429	0.514	0.6	0.686	0.771	0.857
	3200	0	0.091	0.183	0.274	0.366	0.457	0.549	0.64	0.731	0.823	0.914
	3500	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1

CAD MODEL PARAMETER INPUT

Section	Point	X	Y	Z
Left Front Linkage	ROH	-0.32	0.68	-0.11
Left Front Linkage	ROCL1	-0.07	0.23	0
Left Front Linkage	ROCL2	-0.29	0.23	0
Left Front Linkage	ROCL3	-0.05	0.23	0.18
Left Front Linkage	ROCL4	-0.29	0.23	0.18
Left Front Linkage	ROCS	-0.22	0.23	0.35
Left Front Linkage	ROLLAS	-0.23	0.4	-0.07
Left Front Linkage	ROLL2U	-0.32	0.66	-0.28
Left Front Linkage	ROLL3U	-0.32	0.64	0.01
Left Front Linkage	YOCS0	-0.4	0.62	-0.06
Steering	ROR1	-0.36	0.2	0.05
Steering	ROR2	-0.36	-0.2	0.05
Steering	ROQZ	-0.54	0	0.31
Steering	ROQ	-0.68	0	0.45
Rear Left Linkage	ROH	-2.04	0.7	-0.05
Rear Left Linkage	ROCL1	-1.69	0.23	0
Rear Left Linkage	ROCL2	-2.01	0.23	0
Rear Left Linkage	ROCS	-1.88	0.39	0.4
Rear Left Linkage	ROLL2U	-1.98	0.47	-0.07
Rear Left Linkage	ROSB	-1.97	0.49	-0.04